

pkgbase in Production: A Practical Overview

Lukas Engelhardt, EuroBSDCon Brussels, 12.09.2026



pkgbase.de

Agenda

- What is this pkgbase stuff?
- Example network setup
- Building your own packages
- Converting existing Hosts
- Patch level updates
- Minor upgrades
- Major upgrades
- Jail creation
- pkgbase upgrade 14 -> 15

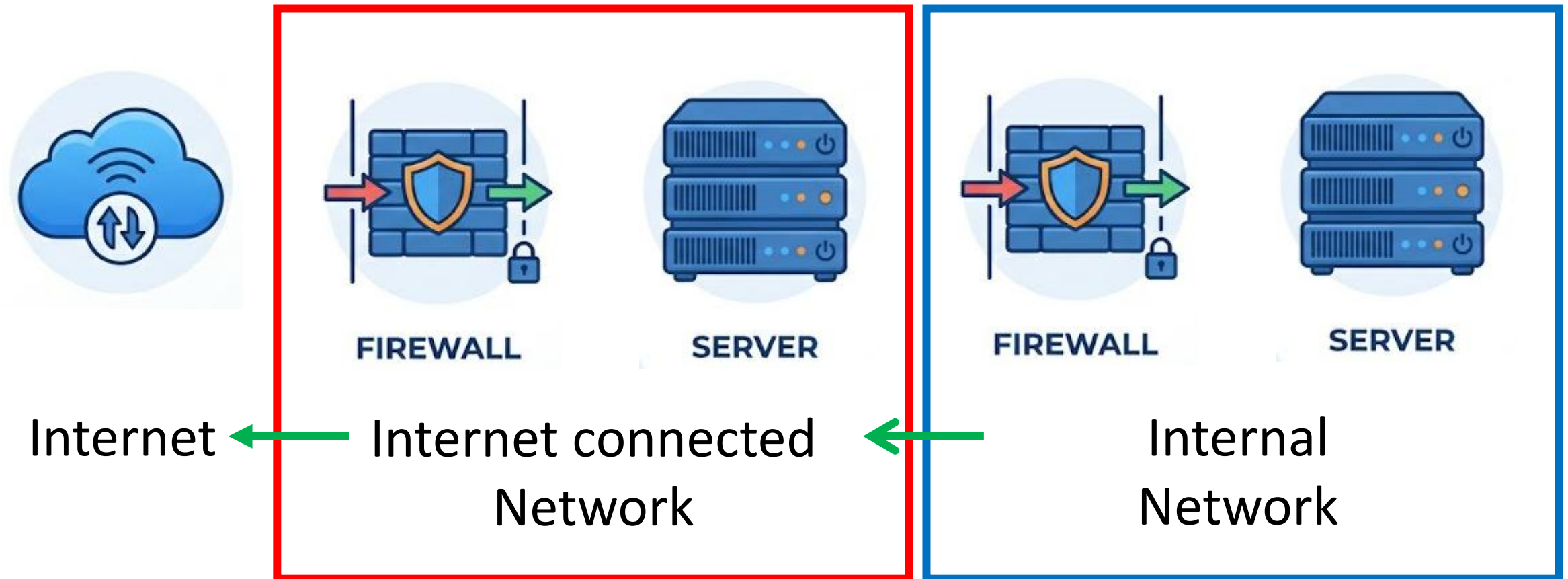
What's that pkgbase stuff?

- pkg(8) now can update your base system
- Base system is packaged in smaller pieces (508 in 15.1 amd64)
- replaces freebsd-update(8) in FreeBSD 16
- FreeBSD-set's allow for minimal installation

FreeBSD-set-*

- minimal
- minimal-jail
- devel
- lib32
- base
- base-jail
- src
- tests
- kernels
- optional

Multi layer network setup



Build your own packages

- without direct internet connection
- for local upgrades
- multiple hosts/jails
- multiple hosts with slow internet?

Build your own packages

- `cd /usr/src`
- `git clone https://github.com/freebsd/freebsd-src.git /usr/src`
- `git checkout release/15.0.0`
- `make -j8 buildkernel && make -j8 buildworld && make -j8 packages`
- `git checkout release/15.0.0-p1`
- `make -j8 buildkernel && make -j8 buildworld && make -j8 update-packages`
- [...] repeat with every patch level

Build your own packages – with poudriere

- `git clone https://github.com/freebsd/freebsd-src.git /usr/src`
- `git checkout release/15.1.0`
- `poudriere jail -c -B -j 15base -m src=/usr/src -b -v 15.1-RELEASE -K GENERIC`
- `git checkout release/15.1.0-p1`
- `poudriere jail -u -b -j 15base`
- [...] repeat with every patch level
- Results in `/usr/local/poudriere/data/images/15pkgbase-repo/FreeBSD:15:amd64/`

Build your own packages – with poudriere

- `poudriere jail -c -j 15base -v release/15.1.0 -B -a amd64 -D -K GENERIC -m git+https`
- `poudriere jail -u -j 15base -t release/15.1.0p1`
- [...] repeat with every patch level

- Results in `/usr/local/poudriere/data/images/15base/FreeBSD:15:amd64/`

- Create poudriere jail without building (this will not give you local packages)

```
poudriere jail -c -j 151amd64 -m pkgbase=base_release_1  
-U https://pkg.freebsd.org/ -v 15
```

Publish your own packages

- `/usr/local/etc/nginx/nginx.conf`

```
server {  
    listen      80;  
    server_name builder;  
    location /  
    {  
        alias /usr/obj/usr/src/repo/;  
        autoindex on;  
    }  
}
```

Converting existing Hosts...

- Update to 15.0 via freebsd-update, install pkg > 2.7+
- Then use pkgbasify!
 - Creates boot environment
 - Creates repo config files
 - Upgrades existing components to pkgbase
 - Merges config files
 - Updates passwd and capabilities databases
 - Restarts sshd
- You are ready!

Demo: patch level update 15.0 -> 15.0p13

- create boot environment
- pkg update
- pkg upgrade

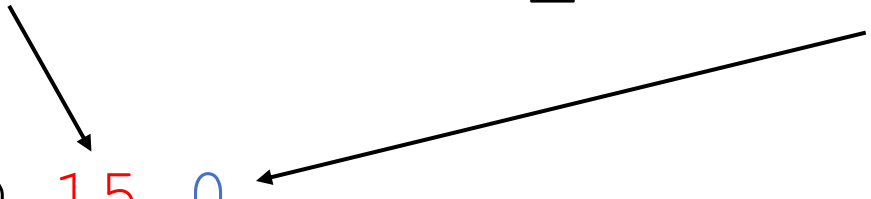
base_release_x explained

```
FreeBSD-base {  
    url =  
    "pkg+https://pkg.freebsd.org/${ABI}/base_release_${VERSION_MINOR}";  
    mirror_type = "srv";  
    signature_type = "fingerprints";  
    fingerprints = "/usr/share/keys/pkg";  
    enabled = yes;  
}
```

base_release_x explained

`pkg.freebsd.org/FreeBSD:15:amd64/base_release_0`

FreeBSD 15.0



FreeBSD 14.4 = ABI FreeBSD:14:amd64, base_release_4

FreeBSD 15.0 = ABI FreeBSD:15:amd64, base_release_0

FreeBSD 15.1 = ABI FreeBSD:15:amd64, base_release_1

FreeBSD 15-CURRENT = ABI FreeBSD:15:amd64, base_latest

FreeBSD 16-CURRENT = ABI FreeBSD:16:amd64, base_latest

Demo: minor upgrades 15.0 -> 15.1

- create boot environment
- change `/usr/local/etc/pkg/repos/FreeBSD-base.conf` url to `base_release_1`
- `pkg upgrade`
- That's it!

Demo: major upgrades

- create boot environment
- `# env ABI=FreeBSD:16:amd64 pkg-static upgrade`
- reboot

Create Jail with base system packages

- `zfs create tank/foo`
- `pkg -r /tank/foo/ install FreeBSD-set-minimal-jail`
- `bsdinstall jail /tank/foo/`
- Create jail config
- Copy resolv.conf in jail
- Start jail

Upgrading from 14 to 15 with pkgbase

*officially
unsupported*

- create boot environment
- mount boot environment
- Use env ABI=FreeBSD:15:amd64
- pkg-static -c /mountpoint install FreeBSD-set-minimal FreeBSD-set-kernels ...
- pkg-static -c /mountpoint remove -fy -g 'FreeBSD-*-14.3*'
- pkg-static -c /mountpoint install FreeBSD-set-minimal FreeBSD-set-kernels ...
- upgrade non-base packages
- Enable boot env
- Change FreeBSD.conf for renamed FreeBSD-ports and FreeBSD-ports-kmods
- Reboot
- Change base config files that have been overwritten

Set minimal + ?

- FreeBSD-set-minimal
- FreeBSD-set-kernels
- FreeBSD-ssh
- FreeBSD-jail
- FreeBSD-ntp
- FreeBSD-vi
- FreeBSD-bsdconfig

Remove base set (just minimal)

- `pkg remove -f FreeBSD-set-base FreeBSD-set-devel FreeBSD-set-lib32
FreeBSD-set-optional FreeBSD-set-src FreeBSD-set-tests`
- `pkg autoremove`
- `pkg install FreeBSD-kernel-generic FreeBSD-ssh FreeBSD-vi FreeBSD-
bsdconfig FreeBSD-jail`

Modify package sets

- An example: add FreeBSD-ssh to the minimal list
- in the freebsd-src repository edit `release/packages/ucl/ssh-all.ucl`

```
annotations {  
    set = "optional,optional-jail,minimal"  
}
```

- Add the package sets you would like to include FreeBSD-ssh in (in this case minimal).
- Build the packages

pkgbase in Production: A Practical Overview

Lukas Engelhardt, EuroBSDCon Brussels, 12.09.2026

develuke@gmx.de